
Can This Agent Even Do That?

A Decidable Goal-Achievability Type Discipline for LLM-Synthesized Agent Skills

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Modern autonomous agents rely on natural-language skills that specify tools,
2 workflows, and desired outcomes. Such skills may be written by human experts or
3 generated with LLM assistance, but their goals may be structurally impossible to
4 achieve: a required tool may be unavailable, a constraint may be unsatisfiable, or
5 agents may deadlock. We present a *skill compiler* that checks goal achievability
6 before execution, without relying on an LLM judge. It combines capability-
7 based reachability from automated planning with direct checking of multiparty
8 communication protocols [1, 2]. An LLM may compact a skill into a formal
9 pack of capabilities, preconditions, effects, goals, and protocol structure, which
10 is then checked deterministically. A Coq-verified abstraction theorem guarantees
11 that an *impossible* verdict is correct for the formal pack being checked. On a
12 deterministic corpus of 15 skills, the checker correctly identifies all 6 achievable and
13 all 7 genuinely impossible cases; the two false *achievable* results are deliberately
14 constructed cases outside the static guarantee. The full evaluation takes less than
15 one second. The checker uses no LLM tokens, while compacting the corpus
16 costs 12.2K tokens compared with approximately 1.07M tokens for one round of
17 corresponding agent executions, an approximately $87\times$ reduction. These results
18 show that a lightweight static check can identify structurally impossible skills
19 before execution at low cost.

20 1 Introduction

21 A modern agent is no longer merely a chatbot, but an autonomous system capable of taking actions
22 beyond text generation—for example, writing code, sending emails, or executing trades. Such
23 behaviour is often guided by natural-language artifacts that specify instructions, workflows, and
24 constraints for the agent [3, 4, 6, 7]. A *skill* is one common way of expressing such guidance. It is a
25 folder containing instructions and tooling that an agent can discover and load as needed [5], thereby
26 *harnessing* a general-purpose agent toward a particular outcome.

27 There is growing evidence that this guidance matters. Across seven state-of-the-art open-source
28 multi-agent systems, failure rates range from 41% to 86.7% [8]. More importantly, some of these
29 failures can be addressed without changing the underlying model. In one system, simply adding a
30 task-objective verification step to the specification improves task success by 15.6% [8, Appendix H].
31 Likewise, repairing the harness based on execution traces improves task completion by 6.3 to 18.4
32 percentage points across four agent benchmarks [4]. These results show that the instructions and
33 structure surrounding an agent can matter as much as the model itself.

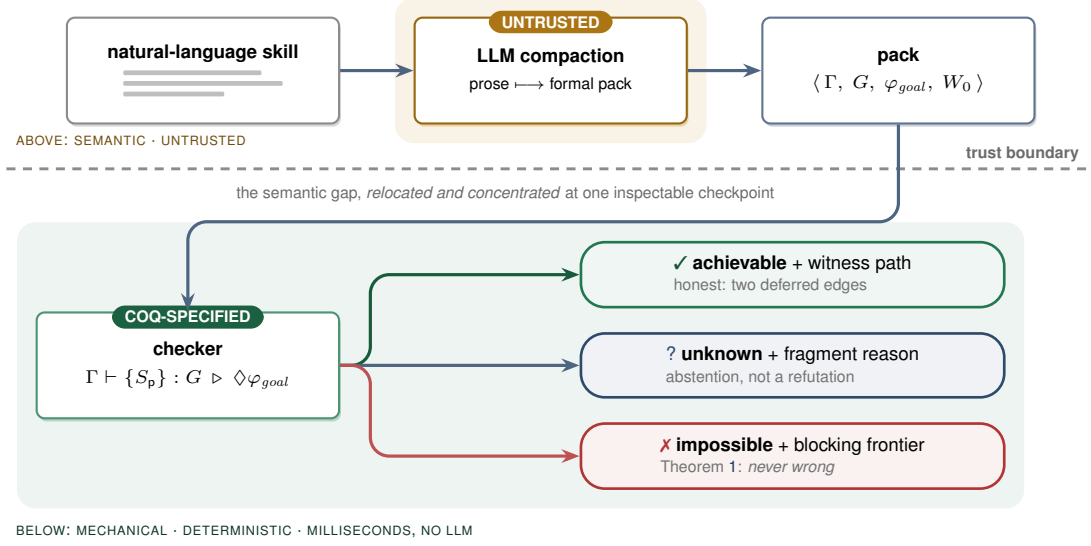


Figure 1: The trust boundary. An untrusted LLM compacts natural-language instructions into a formal pack. A deterministic checker then decides achievability over the pack using a Coq-verified abstraction theorem. The verdicts are asymmetric: *impossible* is sound (Theorem 1), while *achievable* depends on the accuracy of the pack (Section 3.1).

34 The question, however, is how we can know whether such guidance is sufficient to achieve the
 35 outcome it declares. A plan may *book a flight and email a confirmation* even though no email tool is
 36 available. It may pursue *a flight under \$500* even though the available booking tool can return only
 37 premium fares. Or a planner may wait for a worker’s result while the worker waits for the planner’s
 38 answer, with each expecting a message that the other will never send. These are not merely execution
 39 failures. In each case, the goal is already unreachable before execution begins.

40 Such failures are structural and recurring, yet today the only way to discover them is often to run
 41 the skill and inspect what went wrong afterwards [4]. By then, the agent may already have spent
 42 substantial time and tokens taking actions, and a multi-agent system may already have become stuck
 43 waiting for communication that will never occur. A simple static check could, in principle, identify
 44 these problems before execution.

45 This paper asks a deliberately narrow question: **given a skill, can its declared goal be achieved**
 46 **at all?** We answer this question with a deterministic formal typing system. We do not verify that
 47 every step is safe or that the agent will behave correctly on every possible input. Instead, we check
 48 *achievability*. The check is tolerant of small details, detours, and payload variation, which should not
 49 by themselves cause a false alarm. At the same time, it is *sound for refutation*: when the compiler
 50 reports that a goal is impossible, that verdict is correct.

51 1.1 The idea, and the trust boundary

52 The architecture, shown in Figure 1, has two parts separated by a trust boundary. First, an LLM
 53 performs *compaction*: it reads the natural-language skill and turns it into a formal *pack* containing
 54 capabilities with preconditions and effects, a goal-marked global protocol, and an initial state. This
 55 step is *untrusted*: the LLM may misunderstand the skill or omit relevant details.

56 A deterministic *checker* then decides achievability using only the resulting pack. The checker is
 57 mechanically proved sound: if it returns *impossible*, then the goal is impossible under the formal pack
 58 it received. If the LLM omits a constraint or capability from the original skill, the checker cannot
 59 recover information that is not present in the pack; it will instead reason about the incomplete pack.
 60 Conversely, if the LLM introduces a constraint that was not in the original skill, the checker may
 61 report the resulting goal as impossible. Thus, our guarantee is about the checker’s verdict with respect
 62 to the formal pack, not about whether the pack perfectly represents the original natural-language skill.

This does not make the compaction step a weakness that we simply ignore. The formal pack provides a structured target for the LLM to extract. Rather than asking the LLM to freely invent a formal model, we give it a predefined logical structure: capabilities have preconditions and effects, goals are explicit, and the protocol follows a fixed syntax. This makes compaction a constrained translation task and gives us a concrete artifact that can be inspected before execution.

Crucially, the gap between *what the user meant* and *what was formalized* cannot be eliminated. We instead make this gap explicit at the compaction step, where the resulting pack can be inspected and checked before execution. When the compaction is accurate, the checker can identify goals that are impossible before the agent spends time and tokens executing the skill. When the compaction is inaccurate, the checker still provides a sound verdict for the formal pack it received. The key design idea is therefore to move achievability checking from runtime execution to an explicit, inspectable checkpoint before execution.

1.2 Contributions

1. **A goal-achievability type discipline** (Section 3 and Appendices A–C) that combines capability-guarded reachability from automated planning with deadlock-freedom from multiparty session types. The discipline checks whole session configurations directly against a goal-marked global type synthesized from the skill. Building on the multiparty session type tradition of Honda, Yoshida, and Carbone [12, 13], and the direct session-checking approach of SMPS [1, 2], it adds capabilities, explicit goals, and a refutation-sound but achievement-incomplete analysis for LLM-drafted skills.
2. **A mechanized soundness specification** (Appendix D) formalized in Coq. We prove a refutation-sound abstraction theorem parameterized by simulation hypotheses, together with tolerance soundness and capability monotonicity: adding capabilities cannot make an achievable goal impossible. We also establish operational correspondence through subject reduction and session fidelity for the direct judgment over a labelled transition system. For single-label communication, we mechanize the cases where other participants continue to act concurrently; the world-changing action cases are proved on paper under explicit side conditions.
3. **A termination boundary** (Appendix D). When the set of participants is fixed, the symbolic search always terminates, and an *impossible* verdict is sound. Exact achievability over unbounded integers is undecidable, so the checker works on a widened abstraction rather than deciding the exact judgment. Likewise, allowing an unbounded number of new participants to be added during execution can make the problem undecidable, because the search is no longer guaranteed to remain finite.
4. **An implementation and evaluation** (Sections 4–5). We implement the checker together with an LLM-compaction front-end and a schema gate, and evaluate it on a corpus covering the main failure modes. The results are consistent with the theoretical guarantees: we observe no false impossibility verdicts, while every false achievability verdict falls into one of the two deferred cases.

2 Overview by Example

Consider a one-agent skill whose goal is *a flight is booked and a confirmation is sent*. The LLM compacts the skill into a formal pack. If no email tool is available, there is no action that can establish `confirmation_sent`.

$$\begin{aligned}
 \varphi_{goal} &= \text{booked} \wedge \text{confirmation_sent} \\
 \Gamma(a) &= \langle \text{pre}_a, \langle \text{Add}_a, \text{Del}_a, \text{Asg}_a, \text{Nd}_a \rangle \rangle \\
 \text{search} &\mapsto \langle \top, \langle \{\text{searched}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{filter} &\mapsto \langle \text{searched}, \langle \{\text{filtered}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{book} &\mapsto \langle \text{filtered}, \langle \{\text{booked}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{email} &\mapsto \langle \text{booked}, \langle \{\text{confirmation_sent}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{dom}(\Gamma_A) &= \{\text{search}, \text{filter}, \text{book}\} \quad \text{dom}(\Gamma_B) = \text{dom}(\Gamma_A) \cup \{\text{email}\}
 \end{aligned}$$

Under STRIPS [17] frame semantics, the checker therefore returns *impossible* and identifies the missing capability. If the email tool is available, the goal is reachable through the sequence `search`.

filter · book · email, which the checker returns as a witness. The same example is mechanized in Appendix D (the instance `FlightInstance`). Importantly, the booking itself remains reachable in the first case; the refutation concerns only the missing capability needed to complete the goal.

Tolerance is also important. A skill should not be rejected simply because an agent sends extra status messages, takes a clarifying step, or uses a different payload. Our analysis therefore allows these variations in three ways: *existential may-reachability*, where one successful path is sufficient; *session subtyping*, which allows safe interface variation; and *goal-relevant abstraction*, which ignores payload details that do not affect the goal. These mechanisms are defined in Appendices A–C.

3 Verification Model and Signals

Declared pack. The checker consumes $P = \langle \Gamma, G, \varphi_{goal}, W_0 \rangle$. Here, Γ contains capability declarations with guards and STRIPS-style effects; G is a goal-marked global protocol (a *global type*; Appendix A); φ_{goal} is the logical goal; and W_0 is the initial abstract world state, described by a constraint $init \in \mathcal{L}$. Any world satisfying this constraint is a possible initial world, written as $W_0 \models init$. The LLM produces P from the natural-language skill. A deterministic schema gate checks that the pack stays within the part of the language the checker can decide: a finite, fixed set of roles, guarded recursion, finite-domain booleans, and quantifier-free linear integer arithmetic. The formal syntax and semantics are given in Appendices A–B.

Verification signals. The checker uses three static signals:

Signal	Role	Output
tool availability and effects	capability reachability	missing capability or witness action
coordination structure	global protocol conformance	protocol mismatch or conformant protocol
goal and cost refinements	logical reachability	blocked guard or model

These signals are checked before execution. Runtime payload checks remain separate obligations and are not assumed by the static checker.

Decision procedure. After the schema gate, the checker (i) rejects undeclared capabilities and goal conditions that cannot be achieved by any available capability; (ii) checks role behaviors against G , requiring exact sender choices while allowing safe receiver-side branch supersets; and (iii) explores existential protocol/world successors from W_0 , using Z3 [18] for guards and refinements. A goal condition that cannot be satisfied blocks the search. If the solver times out or the pack falls outside these restrictions, the checker returns UNKNOWN rather than *impossible*. The checker therefore returns:

$IMPOSSIBLE(F)$ a sound explanation of why the goal cannot be reached,
 $ACHIEVABLE(\pi, O)$ a witness path π and deferred obligations O ,
 $UNKNOWN(r)$ an abstention with a reason r .

Each verdict records the pack digest, checker semantics, and artifact version.

Guarantees. Let a configuration contain both a protocol state and an abstract world state, and let abs map concrete configurations to abstract ones. Under the step and goal simulation assumptions for the declared pack (Appendix D), if the abstract search cannot reach φ_{goal} , then no concrete execution represented by that pack can reach the concrete goal (Theorem 1). This guarantee remains valid when the abstraction contains less detail, and adding more capabilities cannot cause an otherwise achievable goal to be reported as *impossible* (Theorems 2–3).

The direct conformance check also satisfies subject reduction and session fidelity (Theorems 4–5). The Coq development covers single-label communication together with concurrent actions by other participants. When two actions both change the world state, their interleaving requires an explicit commutativity assumption. The restrictions above ensure that the search terminates. An *impossible* verdict is sound for the declared pack, while an *achievable* verdict only states that the goal is reachable in the abstract search (Proposition 1). Allowing an unbounded number of new participants to be added during execution is an undecidable extension, so the implementation returns UNKNOWN in such cases. Full rules, assumptions, proofs, and mechanization scope appear in Appendix D.

3.1 Structured feedback and human oversight

The checker returns an explanation for IMPOSSIBLE and a witness path for ACHIEVABLE. These outputs provide concrete feedback rather than a single score. A bounded repair adapter may ask the LLM to fix a protocol, but it cannot invent capabilities or weaken the goal. Every revision is checked again by the deterministic checker. Human review remains necessary for the gap between the original intent and the formal pack. Automated prompt or scaffold search is an application of these verdicts, not an empirical contribution of this paper.

4 Implementation

We implemented the checker and its supporting components in Python. The compaction front-end sends the skill to an LLM and passes the result through a deterministic *schema gate* before the checker processes it. Packs that do not satisfy the schema are rejected at this stage. The checker implements the abstractions from Appendices A–C: STRIPS frame semantics for the world, existential search over the protocol, and Z3 for guard and goal satisfiability and arithmetic refinements. On success it returns a witness path; on refutation, it reports the reason, such as a missing capability, unsatisfiable guard, unachievable goal condition, or non-projectable handoff. The optional LLM front-end may use a NON_PROJECTABLE counterexample for one bounded repair round. It may not invent capabilities or weaken the goal, and the deterministic checker remains the final decision maker.

The conformance step ($G \vdash (\mathbb{M}; W)$, Section C.2) is implemented by means of a simple *type inference algorithm* since all typing rules are structural in $(\mathbb{M}; W)$. At each step we freely choose either one or two participant processes to reduce or a goal satisfied by the world obtaining the set of sessions which must be typed as premises of the typing rule.

The adapter also checks the participant side conditions of T-COMM/T-ACT/T-GOAL: it computes $\text{prt}(G)$ and compares it with the roles for which the pack declares behaviour. A declared role outside $\text{prt}(G)$ has contract End, so a non-trivial behavior for that role cannot conform. If the pack declares no behavior for a participant, the checker reports the missing role instead of leaving the assumption implicit.

5 Evaluation

We assembled a corpus of 15 skills covering the main failure modes: hallucinated planning, retry-forever loops, coordination deadlock, and refinement violations. The corpus also includes achievable controls with detours and informed branches, plus two deliberately *spurious* cases for the deferred checks. Each skill has a ground-truth semantic label; the positive class is *achievable*. Evaluation is deterministic and CPU-only: it requires no training or live model inference, completes the 15-case matrix in under one second on a commodity laptop, and gives each individual Z3 query a 10-second timeout.

	predicted achievable	predicted impossible
truly achievable	TP = 6	FN = 0
truly impossible	FP = 2	TN = 7

Two claims, both confirmed. **Soundness:** FN = 0 — no achievable goal was wrongly refuted, consistent with Theorem 1 (achievability recall $6/6 = 100\%$). **Incompleteness, contained:** FP = 2, and both are the planted spurious cases — one a false payload post-condition (Layer-C obligation), and one an under-specified goal (intent edge). No structural failure was missed in this proof-of-concept corpus. This is evidence of coverage, not a proof that these are the only possible failure modes. Each refutation reason matches its intended failure mode:

Failure mode (source)	Verdict reason
hallucinated planning — invokes a non-existent tool	MISSING_CAPABILITY
hallucinated planning — no establisher for a goal conjunct	GOAL_UNSAT
retry-forever loop — guard never satisfiable	BLOCKED_GUARD
coordination deadlock — unobserved choice / bad handoff	NON_PROJECTABLE
refinement violation — budget unsatisfiable on every run	GOAL_UNSAT

What the check costs. The checker and deterministic front-end use *zero* LLM tokens. The LLM is used once to compact each skill *version*, while a full agent run uses tokens on every *invocation*. Under the conservative model in Appendix E, compacting the 15-spec corpus costs 12.2K tokens and avoids $\sim 1.07\text{M}$ tokens per round of invocations, giving $\sim 87\times$ leverage. For every failure mode, the cost of compaction is recovered by the first prevented run. For a skill that is *not* broken, the check costs 2.9% of one successful run, or nothing.

A separate six-case suite tests behavior not covered by the main matrix: tail-recursive retry and non-terminating control, adding new participants during execution, refutation that remains valid when new participants are added, and conformant versus non-conformant declared role behavior. It yields two ACHIEVABLE, three IMPOSSIBLE, and one UNKNOWN. The abstention is reported separately rather than counted as a false negative because UNKNOWN makes no refutation claim. Together the suites contain 21 declared packs; the 15-case matrix remains the only binary ground-truth matrix.

6 Related Work

Multiparty session types [13, 16, 9] provide formal methods for checking communication protocols, including deadlock-freedom through projection and subtyping [14, 10]. Our protocol follows the synchronous formulation of this line of work [9–11]. We also follow the recent approach of checking a whole session configuration directly against a global type [1, 2], rather than first projecting the global type to local types. The new contribution is the world and goal layer: capability guards, effects, goal observation, and the asymmetric refutation guarantee are not modelled by classical MPST or new SMPS. These are not part of classical MPST [13, 16, 9] or the recent direct-checking formulation [1, 2].

Automated planning [17] provides methods for deciding whether a goal can be reached from an initial state using available actions. We use the same basic idea of preconditions and effects, but apply it to agent skills and combine it with multiparty communication protocols. The resulting checker asks whether the declared goal can be achieved given both the available capabilities and the communication protocol.

Recent work on agent verification addresses related problems but has different goals. *Provable coordination via message sequence charts* [20] uses formal communication protocols for LLM agents and focuses on coordination correctness rather than goal achievability. Its runtime-planning extension also studies LLM-generated coordination protocols. *VeriGuard* [21] separates offline policy verification from online monitoring and uses the result as a safety mechanism. *Agent behavioural contracts* [22] provide probabilistic, runtime-enforced compliance and composition conditions for multi-agent systems. In contrast, our checker makes a static achievability decision and guarantees that an *impossible* verdict is correct for the formal pack.

Skill-bundle scanners and verified-skill catalogs address the same deployment object: portable agent skills. NVIDIA’s SkillSpector [23] and Verified Agent Skills pipeline [24] provide checks for skill bundles before deployment, including code, metadata, dependencies, and other security-related properties. These checks are complementary to our approach. They can inspect the original SKILL.md and its supporting files, while our checker operates on the formal pack produced from the skill. Their purpose is to identify security and consistency problems; ours is to determine whether the declared goal can be achieved with the available capabilities and protocol.

To our knowledge, the work above does not combine capability-based goal reachability with multiparty protocol checking for agent skills, nor provide the same refutation-sound guarantee for an *impossible* verdict.

7 Limitations and Future Work

- **Dependence on the declared pack.** The guarantees apply to the declared Γ and S . If the LLM incorrectly describes a capability or omits an important detail from the skill, the checker can only reason about the resulting pack. A practical system should therefore also inspect SKILL.md, scripts, dependencies, manifests, and permission metadata before running the achievability check.

246 • **Adding participants during execution.** The current procedure terminates when the set of
 247 participants is fixed. When new participants can be added during execution, the procedure
 248 may return UNKNOWN rather than make an unsupported decision (Proposition 2).

249 • **Abstraction.** A coarse α allows more behaviors to be treated as equivalent. This preserves
 250 the refutation guarantee of Theorem 1, but may make the checker more likely to return
 251 *achievable*.

252 • **Formalization gaps.** The Coq development covers the generic refutation-sound abstraction
 253 theorem and concrete examples, head-move action safety, and subject reduction/session
 254 fidelity with full bystander interleaving for the single-label communication model, all axiom-
 255 free. The executable checker is schema-gated and tested against the formal specification,
 256 but its exact symbolic state is not yet fully mechanized (Lemma 3 is on paper). The
 257 Coq model also does not yet combine participant matching, multi-branch choice, and
 258 world-changing interleaving in one development. Future work includes completing this
 259 mechanization, connecting interleaved session runs to the reachability search, proving
 260 equivalence between the declarative judgment and the projection-based adapter, and proving
 261 that abstract witnesses correspond to concrete sessions. The 21-pack evaluation is also only
 262 a proof of concept.

263 **Broader impact.** Pre-execution checking can reduce wasted computation (Appendix E) and prevent
 264 structurally impossible plans from reaching execution. However, an incorrect pack can create false
 265 confidence, for example by omitting a tool effect or weakening the user’s actual goal. We therefore
 266 report the formal pack, the result of the check, and any remaining conditions that still require attention.
 267 Human review and runtime checks are still needed for consequential use. This work does not evaluate
 268 fairness, privacy, or deployment safety.

269 8 Conclusion

270 Agent skills provide instructions, workflows, and tools that guide autonomous agents toward particular
 271 outcomes. Whether a skill is written by a human expert or generated with the help of an LLM, a basic
 272 question remains: **can the declared goal be achieved at all?** Our work answers this question before
 273 execution, using a deterministic checker over a formal pack derived from the skill. The checker
 274 combines capability-based reachability with a directly checked global protocol, and its *impossible*
 275 verdict is sound for the formal pack it receives.

276 This guarantee does not remove the gap between the original skill and its formal representation. An
 277 incorrect or incomplete pack can lead to an incorrect *achievable* result, which is why the result must
 278 be understood relative to the declared pack. Nevertheless, when the pack is correct, the check can
 279 identify structural failures before the agent spends time and tokens executing the skill. This provides
 280 a simple and inspectable way to check whether an agent skill is achievable before execution.

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A Syntax

We use disjoint sets of predicates $p \in \text{Pred}$ (boolean), numeric variables $x \in \text{Var}$, roles $p, q, r \in \mathcal{R}$, capability names $a \in \text{Cap}$, and message labels $\ell \in \text{Lab}$. Following the modern presentation of multiparty sessions [1, 2] we define the model from the bottom up: state and capabilities (Sections A.1–A.2), then *processes* (Section A.3), and finally the global *type* they are checked against directly (Section A.4).

In each grammar, “ $::=$ ” lists the possible forms of an expression. In each inference rule below, the statement below the line holds when all premises above the line hold. Rule names identify the corresponding checker obligation; they are not additional assumptions.

337 A.1 State logic

$$\begin{aligned} e &::= x \mid n \mid e + e \mid e - e \mid c \cdot e & (c, n \in \mathbb{Z}) \\ \varphi &::= p \mid \top \mid \perp \mid \neg \varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid e \bowtie e & \bowtie \in \{<, \leq, =, \geq, >, \neq\} \end{aligned}$$

338 The state logic $\mathcal{L}$ is quantifier-free linear integer arithmetic [19] with uninterpreted boolean predicates
 339 — a decidable fragment for which entailment $\models$ and satisfiability are discharged by a satisfiability-
 340 modulo-theories (SMT) solver.

341 A.2 Worlds and capabilities

342 A *world* $W = \langle B, N \rangle$ assigns truth values to predicates $B : \text{Pred} \rightarrow \mathbb{B}$ and numeric variables
 343 $N : \text{Var} \rightarrow \mathbb{Z}$. A *capability context* Γ maps each available tool a to a guarded effect:

$$\begin{aligned} \Gamma(a) &= \langle pre_a, eff_a \rangle, \quad pre_a \in \mathcal{L}, \\ eff_a &= \langle Add_a \subseteq \text{Pred}, Del_a \subseteq \text{Pred}, Asg_a : \text{Var} \rightarrow e, Nd_a : \text{Var} \rightarrow \mathcal{L} \rangle \end{aligned}$$

344 $a \notin \text{dom}(\Gamma) \iff$ the agent does *not* possess tool a (no hallucinated tools).

345 Add/Del are the STRIPS add- and delete-lists (well-formedness requires $Add_a \cap Del_a = \emptyset$, so the
 346 order of union and difference in apply below is immaterial); Asg gives deterministic numeric updates
 347 $x := e$; Nd gives *nondeterministic* updates $x := *$ constrained by a formula over the new value (used
 348 to model post-conditions such as “books a fare under 500”).

349 A.3 Processes and multiparty sessions

350 Skills *execute* as processes; a multi-agent run is a session of named processes. We use the synchronous
 351 multiparty session calculus in its modern coinductive presentation [1, 2]: processes coinductively
 352 defined are possibly infinite *regular* trees (finitely many distinct subtrees), and labels are goal-relevant
 353 tokens, $\ell = \alpha(m)$ for concrete messages m — the tolerance dial α is a parameter of the label
 354 alphabet.

$$P ::=^{coind} \mathbf{0} \mid p!\{\ell_i.P_i\}_{i \in I} \mid p?\{\ell_i.P_i\}_{i \in I} \mid a.P \quad \mathbb{M} = p_1[P_1] \parallel \dots \parallel p_n[P_n]$$

355 with I finite and non-empty, labels ℓ_i pairwise distinct, and role names pairwise distinct. The output
 356 $p!\{\ell_i.P_i\}$ *internally* chooses a label and sends it to p ; the input $p?\{\ell_i.P_i\}$ offers an *external* choice;
 357 $a.P$ invokes capability a — the only construct that changes the world. A skill S_p is a process: the
 358 behaviour declared for role p .

359 A.4 Global types

360 The single communication construct of a global type is a *directed* labelled choice — selection by p
 361 and branching at q in one construct. (A partnerless “ p selects” admits no coherent view for the roles
 362 that did not choose; directed choice is the standard MPST construct [13, 9].)

$$G ::=^{coind} p \rightarrow q : \{\ell_i.G_i\}_{i \in I} \mid a@p.G \mid \check{\varphi}.G \mid \text{End}$$

363 subject to $p \neq q$ and the same regularity and label conditions as processes. This is the *only* type
 364 in the discipline: Section C.2 types a whole session directly against G — there is no local-type
 365 grammar, no projection function, and no separate subtyping relation. Goal markers $\check{\varphi}$ live *only*
 366 in G : achievability is decided on the global type (Rule ACH in C.3), and markers are consumed at
 367 typing time by T-GOAL without ever burdening a process. Only $a@p$ changes the world; messages
 368 carry coordination information only.

369 B Operational Semantics

370 B.1 World transitions

A capability fires when its guard holds, producing a successor world. Because Nd is nondeterministic,
 $\langle W \rangle a \langle W' \rangle$ may relate one world to many. The relation is implicitly indexed by Γ : the notation
 pre_a, eff_a is defined only when $\Gamma(a) = \langle pre_a, eff_a \rangle$, so every firing presupposes $a \in \text{dom}(\Gamma)$.

$$\frac{\text{WORLD-ACT} \quad W \models pre_a \quad W' \in \text{apply}(eff_a, W)}{\langle W \rangle a \langle W' \rangle}$$

371

$$\text{apply}(\text{eff}_a, \langle B, N \rangle) = \langle B', N' \rangle, \quad B' = (B \cup \text{Add}_a) \setminus \text{Del}_a$$

372

$$N'(x) = \begin{cases} \llbracket \text{Asg}_a(x) \rrbracket_N & \text{if } x \in \text{dom}(\text{Asg}_a) \\ v \text{ with } \langle B, N[x \mapsto v] \rangle \models \text{Nd}_a(x) & \text{if } x \in \text{dom}(\text{Nd}_a) \\ N(x) & \text{otherwise (frame)} \end{cases}$$

373 $\llbracket e \rrbracket_N$ is the usual integer evaluation of e , reading each variable from N . When several variables are in
 374 $\text{dom}(\text{Nd}_a)$, each constraint is evaluated with only its own variable updated; the schema gate rejects
 375 an $\text{Nd}_a(x)$ that mentions another updated variable.

376 B.2 A labelled semantics for sessions and global types

377 Only capabilities touch the world, so sessions reduce as configurations $(\mathbb{M}; W)$; communication is
 378 synchronous [10]. Each transition carries a label Λ recording its *participants*, so that independent
 379 moves by disjoint roles can be identified. Two auxiliary maps drive the labelled system: the
 380 *participants* $\text{prt}(\cdot)$ of a global type, a label, or a session, and the *capabilities* $\text{cap}(\cdot)$ (the set of moves
 381 a global type can perform). On global types,

$$\begin{aligned} \text{prt}(\text{p} \rightarrow \text{q} : \{\ell_i. G_i\}_{i \in I}) &= \{\text{p}, \text{q}\} \cup \bigcup_{i \in I} \text{prt}(G_i) & \text{prt}(\checkmark \varphi. G) &= \text{prt}(G) \\ \text{prt}(a @ \text{p}. G) &= \{\text{p}\} \cup \text{prt}(G) & \text{prt}(\text{End}) &= \emptyset, \end{aligned}$$

382 on labels $\text{prt}(\text{p} \ell \text{q}) = \{\text{p}, \text{q}\}$, $\text{prt}(a @ \text{p}) = \{\text{p}\}$, $\text{prt}(\checkmark \varphi) = \emptyset$, and on sessions $\text{prt}(\mathbb{M}) = \{\text{p} \mid \mathbb{M} \equiv$
 383 $\text{p}[P] \ \& \ P \neq \mathbf{0}\}$ (the roles with a residual behaviour). The capabilities of a global type are

$$\begin{aligned} \text{cap}(\text{p} \rightarrow \text{q} : \{\ell_i. G_i\}_{i \in I}) &= \{\text{p} \ell_i \text{q} \mid i \in I\} \cup \bigcup_{i \in I} \text{cap}(G_i) \\ \text{cap}(a @ \text{p}. G) &= \{a @ \text{p}\} \cup \text{cap}(G) & \text{cap}(\checkmark \varphi. G) &= \{\checkmark \varphi\} \cup \text{cap}(G) & \text{cap}(\text{End}) &= \emptyset. \end{aligned}$$

384 The capability names used by a protocol are

$$\text{caps}(G) = \{a \mid \exists \text{p}. a @ \text{p} \in \text{cap}(G)\}.$$

The LTS of sessions is defined by

$$\begin{array}{c} \text{S-COMM} \\ \hline \frac{k \in I \subseteq J}{(\text{p}[\text{q}!\{\ell_i. P_i\}_{i \in I}] \parallel \text{q}[\text{p}?\{\ell_j. Q_j\}_{j \in J}] \parallel \mathbb{M}; W) \xrightarrow{\text{p} \ell_k \text{q}} (\text{p}[P_k] \parallel \text{q}[Q_k] \parallel \mathbb{M}; W)} \\ \\ \text{S-ACT} \\ \hline \frac{\langle W \rangle a \langle W' \rangle}{(\text{p}[a.P] \parallel \mathbb{M}; W) \xrightarrow{a @ \text{p}} (\text{p}[P] \parallel \mathbb{M}; W')} \end{array}$$

385 where $\parallel$ is commutative, associative and $\text{p}[\mathbf{0}] \parallel \mathbb{M} \equiv \mathbb{M}$. A non-terminated configuration with
 386 no successor is *stuck*. The planner/worker deadlock from Section 1 is one example: both processes
 387 wait for an input, so neither can take the next step. A goal marker is not part of any process, and
 388 satisfying the goal does not create a session transition. Instead, the predicate $\text{goal}_{\varphi_{\text{goal}}}$ below checks
 389 whether the goal holds. Session labels Λ therefore contain only communications and actions, as in
 390 the Coq development. On the type side, the goal marker is discharged together with the next step by
 391 G-GOAL-F below.

The checker relates a session to its protocol through a *labelled* transition system on global types,
 $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, built from *head* rules that consume the leading construct and *interleaving*

rules that let a move by participants *disjoint* from the head commute inward:

$$\begin{array}{c}
\text{G-COMM-E} \quad \frac{k \in I}{(\mathbf{p} \rightarrow \mathbf{q}; \{\ell_i.G_i\}_{i \in I}, W) \rightsquigarrow_{\mathbf{p} \ell_k \mathbf{q}} (G_k, W)} \quad \text{G-ACT-E} \quad \frac{\langle W \rangle a \langle W' \rangle}{(a @ \mathbf{p}. G, W) \rightsquigarrow_{a @ \mathbf{p}} (G, W')} \\
\\
\text{G-GOAL-E} \quad \frac{W \models \varphi}{(\check{\varphi}. G, W) \rightsquigarrow_{\check{\varphi}} (G, W)} \\
\\
\text{G-COMM-I} \quad \frac{\forall i \in I. ((G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W') \wedge \Lambda \in \text{cap}(G_i)) \quad \text{prt}(\Lambda) \cap \{\mathbf{p}, \mathbf{q}\} = \emptyset}{(\mathbf{p} \rightarrow \mathbf{q}; \{\ell_i.G_i\}_{i \in I}, W) \rightsquigarrow_{\Lambda} (\mathbf{p} \rightarrow \mathbf{q}; \{\ell_i.G'_i\}_{i \in I}, W')} \\
\\
\text{G-ACT-I} \quad \frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G) \quad \text{prt}(\Lambda) \cap \{\mathbf{p}\} = \emptyset}{(a @ \mathbf{p}. G, W) \rightsquigarrow_{\Lambda} (a @ \mathbf{p}. G', W')} \\
\\
\text{G-GOAL-F} \quad \frac{W \models \varphi \quad (G, W) \rightsquigarrow_{\Lambda} (G', W')}{(\check{\varphi}. G, W) \rightsquigarrow_{\Lambda} (G', W')}
\end{array}$$

392 In G-COMM-I, all branches take the same Λ and reach the same W' . For communication, $W' = W$,
 393 which is the only case needed here. G-GOAL-E discharges a goal marker when $W \models \varphi$. In the
 394 achievability search, an unsatisfied goal marker blocks that path. In the labelled correspondence,
 395 G-GOAL-F discharges the marker and takes the next step in the same transition, as in GS_Goal in the
 396 Coq development. The goal is checked in the world state where the step occurs, so no goal-stability
 397 assumption is needed. The goal marker $\check{\varphi}$ is not a label in the correspondence.

398 The extraction (head) rules G-COMM-E/G-ACT-E/G-GOAL-E give the Λ -erased reduction
 399 $(G, W) \rightsquigarrow (G', W')$ that the achievability judgment below uses; the interleaving rules matter
 400 only for the operational correspondence of Section D.3, and their $\Lambda \in \text{cap}(G)$ premise confines
 401 a commuting move to a capability the residual protocol actually offers. G-COMM-E is legitimate
 402 because choice is directed — the branch is a communication both endpoints observe, and Section C.2
 403 checks that every third party can follow. A well-formed pack uses its declared goal φ_{goal} at each goal
 404 marker. A configuration is *goal-satisfying* when control is at such a marker or at the terminus and the
 405 world entails that declared goal:

$$\begin{array}{c}
\text{406} \quad \text{goal}_{\varphi_{\text{goal}}}(\check{\varphi}_{\varphi_{\text{goal}}}. G, W) \iff W \models \varphi_{\text{goal}}, \quad \text{goal}_{\varphi_{\text{goal}}}(\text{End}, W) \iff W \models \varphi_{\text{goal}}, \\
\Gamma; G \models \Diamond_{\text{init}} \varphi_{\text{goal}} \iff \exists W_0 \models \text{init}. \exists (G', W'). (G, W_0) \rightsquigarrow^* (G', W') \wedge \text{goal}_{\varphi_{\text{goal}}}(G', W').
\end{array}$$

407 The diamond is *existential*: a single witnessing run suffices — the formal source of detour-tolerance.

408 B.3 Realizability and subtyping, without projection

409 The classical route to ruling out a deadlocking handoff is *projection*: compute each role's local
 410 type $G \upharpoonright r$ by structural recursion on G , taking the *merge* of a choice's branches at every role that
 411 neither sends nor receives, and declaring G *realizable* exactly when this partial function is defined
 412 everywhere — undefinedness of the merge is the static signature of a role forced to behave differently
 413 in branches it cannot distinguish. Session subtyping $\leq$ is then a second, separate coinductive relation
 414 on local types, letting a role's actual behaviour offer more external choices and fewer internal ones
 415 than its projected contract demands [14, 10, 11].

416 Both devices exist to answer questions about the *network as a whole* — “can every role tell the
 417 branches apart it needs to?”, “may this implementation stand in for that contract?” — while type-
 418 checking only one role's syntax at a time. Section C.2 answers the same two questions with a single
 419 judgment, *typing the whole configuration directly against G*: a choice node's rule quantifies over
 420 every branch the protocol declares and requires the *same* residual configuration to check against each
 421 one. This is the conservative plain-merge condition, obtained without ever computing $\sqcap$ or a local
 422 type. The receiver-side subtyping direction (a receiver may accept a superset of the protocol's labels)

is folded into the same rule as a label-set inclusion $I \subseteq J$. Tolerance therefore still has the same three independent knobs from Section 2 — the existential $\diamond$ (Section B.2), subtyping (now inside T-COMM, Section C.2), and the granularity of α — “flexible, tolerant of small details” is still the profile $\langle \text{coarse } \alpha, \diamond, \text{linear } \mathcal{L} \rangle$; a later safety layer flips the same machinery to the universal $\square$ over permission predicates.

C The Achievability Type System

Achievability now factors into three premises, each checked by a separate terminating procedure, combined by the top rule ACH: no hallucinated capability, direct conformance of the declared skills to G , and reachability of the goal.

C.1 Capability and goal typing

$$\frac{\text{T-EFF} \quad \Gamma(a) = \langle pre, eff \rangle \quad \varphi \models pre}{\Gamma \vdash \{\varphi\} a \{\text{sp}(\varphi, eff)\}} \quad \text{T-MISS} \quad \frac{a \notin \text{dom}(\Gamma)}{\Gamma \vdash a \triangleright \mathbf{X}(\text{MISSING_CAPABILITY})}$$

$\text{sp}(\varphi, eff)$ is the strongest postcondition of the STRIPS effect. T-MISS fires on hallucinated planning: the plan names a tool the context does not grant, and achievability is refuted without any search.

C.2 Direct conformance typing

Skills are typed *directly* against the global protocol and the world, by the coinductive judgment $G \vdash (\mathbb{M}; W)$ — read “ G types session $\mathbb{M}$ starting from world W .” There is no local type, no projection, and no separate subtyping relation: the receiver-side subtyping direction and the unobserved-choice check are structural side conditions of one rule.

$$\begin{array}{c} \text{T-END} \\ \hline \text{End} \vdash (\mathbf{p}[0]; W) \\ \\ \text{T-ACT} \\ \frac{\forall W'. \langle W \rangle a \langle W' \rangle \implies G \vdash (\mathbf{p}[P] \parallel \mathbb{M}; W') \quad \text{prt}(G) \cup \{\mathbf{p}\} = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}\}}{a @ \mathbf{p}. G \vdash (\mathbf{p}[a.P] \parallel \mathbb{M}; W)} \\ \\ \text{T-GOAL} \\ \frac{G \vdash (\mathbb{M}; W) \quad W \models \varphi \quad \text{prt}(G) = \text{prt}(\mathbb{M})}{\checkmark \varphi. G \vdash (\mathbb{M}; W)} \\ \\ \text{T-COMM} \\ \frac{G_i \vdash (\mathbf{p}[P_i] \parallel \mathbf{q}[Q_i] \parallel \mathbb{M}; W) \quad \forall i \in I \subseteq J \quad \text{prt}(\mathbf{p} \rightarrow \mathbf{q} : \{\ell_i.G_i\}_{i \in I}) = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}, \mathbf{q}\}}{\mathbf{p} \rightarrow \mathbf{q} : \{\ell_i.G_i\}_{i \in I} \vdash (\mathbf{p}[\mathbf{q}!\{\ell_i.P_i\}_{i \in I}] \parallel \mathbf{q}[\mathbf{p}?\{\ell_j.Q_j\}_{j \in J}] \parallel \mathbb{M}; W)} \end{array}$$

T-ACT checks the protocol and world state together with WORLD-ACT. The type $a @ \mathbf{p}. G$ is checked before the action in W , and the remaining type G is checked after the action in W' — the same convention as G-ACT-E (Section B.2). The second premise considers every possible successor world because an Nd effect may produce different outcomes. Subject reduction therefore requires the remaining type to be valid in each successor. For capabilities without Nd , there is only one successor, so the rule is unchanged.

In T-COMM the sender $\mathbf{p}$ offers exactly the protocol’s branch set I , while the receiver $\mathbf{q}$ may accept *more* ($I \subseteq J$) — the one subtyping direction the rule keeps (safe interface slack at the receiver). Requiring *every* protocol branch ℓ_i to be checked against the *same* bystanders $\mathbb{M}$ is the plain-merge form of the unobserved-choice condition, so a role forced to behave differently across branches it cannot distinguish makes no derivation possible — the rule fails closed, and deadlock-freedom falls out of T-COMM itself with no merge to compute (Theorem 4). Each rule additionally carries a *participant-matching* side condition tying the protocol’s participants to the session’s: $\text{prt}(\mathbf{p} \rightarrow \mathbf{q} : \{\ell_i.G_i\}) = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}, \mathbf{q}\}$ in T-COMM, $\text{prt}(G) \cup \{\mathbf{p}\} = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}\}$ in T-ACT, and $\text{prt}(G) = \text{prt}(\mathbb{M})$ in T-GOAL. These make the invariant “the roles the protocol still

mentions are exactly the roles still running” hold at every node, ruling out a session that carries a stray participant the protocol has forgotten (or vice versa) — the well-formedness that lets the interleaving cases of Section D.3 match a bystander move on the session to a $\text{cap}(\cdot)$ -labelled move on the type. T-END closes the discipline at the terminus: the terminated protocol End types a single idle role $p[0]$ — and, through the congruence $p[0] \parallel \mathbb{M} \equiv \mathbb{M}$, any session all of whose roles have terminated — the base case of the participant invariant, since $\text{prt}(\text{End}) = \emptyset$ and an idle role contributes nothing to $\text{prt}(\mathbb{M})$. T-ACT is derivable only when $\langle W \rangle a \langle W' \rangle$ holds, which itself presupposes $a \in \text{dom}(\Gamma)$: this is where hallucinated capabilities die, with T-MISS as the diagnostic.

C.3 The achievability judgment

$$\text{ACH} \quad \frac{\text{caps}(G) \subseteq \text{dom}(\Gamma) \quad \exists W_0 \models \text{init}. G \vdash (p_1[S_{p_1}] \parallel \dots \parallel p_n[S_{p_n}]; W_0) \wedge \Gamma; (G, W_0) \models \Diamond \varphi_{\text{goal}}}{\Gamma \vdash \{S_p\}_p : G \triangleright \Diamond \varphi_{\text{goal}}}$$

where $\{p_1, \dots, p_n\} = \text{prt}(G)$: the judgment takes one declared skill S_p per role of the protocol. Read top-to-bottom: no hallucinated tools; some plausible initial world exists from which the declared skills directly conform to the protocol (deadlock-free, every capability call guarded) *and* that same world reaches the goal. The reachability half is unchanged from Section B.2 and decided on Γ, G alone, before any S_p is known — exactly what lets the checker run on the LLM-compacted pack the moment it exists; conformance is the additional, separately checked premise once the skills are declared, and needs no transport lemma: the receiver-side subtyping slack is already inside T-COMM.

C.4 Reachability rules

The diamond is derived by the following inductive system, realized by the checker as a finite forward search over $\rightsquigarrow$; unlike T-COMM/T-ACT/T-GOAL, it never mentions a session $\mathbb{M}$, which is what makes it checkable on the pack alone.

$$\begin{array}{c} \text{REACH-GOAL} \\ \frac{W \models \varphi_{\text{goal}} \quad G = \text{End} \vee \exists G'. G = \checkmark \varphi_{\text{goal}}. G'}{\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}} \\[10pt] \begin{array}{cc} \text{REACH-STEP} & \text{REACH-OR} \\ \frac{(G, W) \rightsquigarrow (G', W') \quad \Gamma; (G', W') \models \Diamond \varphi_{\text{goal}}}{\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}} & \frac{\exists k \in I. \Gamma; (G_k, W) \models \Diamond \varphi_{\text{goal}}}{\Gamma; (p \rightarrow q: \{\ell_i.G_i\}_{i \in I}, W) \models \Diamond \varphi_{\text{goal}}} \end{array} \end{array}$$

REACH-OR is derivable from REACH-STEP with G-COMM-E; we keep it because the search implements branching this way.

D Metatheory

We separate what is *mechanized in Coq* (Theorems 1–3; Theorems 4–5 for the single-label communication fragment with full interleaving; the head-move action case; and the concrete instances, all axiom-free) from what is *proved on paper* (the action-interleaving cases, Lemma 3, termination, and the undecidability boundary).

D.1 Soundness of the abstraction

Let $\mathcal{C}$ be the concrete configuration space of protocol/world pairs, $\mathcal{A}$ the abstract configuration space the checker explores, and $\text{abs} : \mathcal{C} \rightarrow \mathcal{A}$ the abstraction. We require:

$$(\text{step-sim}) \quad C \rightsquigarrow C' \Rightarrow \text{abs}(C) \rightsquigarrow^* \text{abs}(C'), \quad (\text{goal-sim}) \quad \text{goal}_{\varphi_{\text{goal}}}(C) \Rightarrow \text{goal}(\text{abs}(C)).$$

Lemma 1 (Transport). *If $C_0 \rightsquigarrow^* C$ then $\text{abs}(C_0) \rightsquigarrow^* \text{abs}(C)$.*

Proof. By induction on the reflexive-transitive closure. The empty run is reflexivity. For $C_0 \rightsquigarrow^* C' \rightsquigarrow C$, the induction hypothesis gives $\text{abs}(C_0) \rightsquigarrow^* \text{abs}(C')$ and STEP-SIM gives $\text{abs}(C') \rightsquigarrow^* \text{abs}(C)$; compose them. Mechanized as `reach_abs`, parameterized by STEP-SIM. $\square$

482 **Lemma 2** (Reachability monotonicity). *If every step of $\rightsquigarrow_1$ is a step of $\rightsquigarrow_2$ (that is, $x \rightsquigarrow_1 y$ implies*
 483 *$x \rightsquigarrow_2 y$), then $s \rightsquigarrow_1^* w$ implies $s \rightsquigarrow_2^* w$.*

484 *Proof.* By induction on the closure $s \rightsquigarrow_1^* w$. The empty run transports by reflexivity. For $s \rightsquigarrow_1^* u \rightsquigarrow_1$
 485 w , the induction hypothesis gives $s \rightsquigarrow_2^* u$; the hypothesis turns the last step $u \rightsquigarrow_1 w$ into $u \rightsquigarrow_2 w$;
 486 appending it gives $s \rightsquigarrow_2^* w$. Mechanized as `reach_mono`. $\square$

487 D.2 Refutation soundness, tolerance, monotonicity

488 **Theorem 1** (Refutation soundness). *If the abstract system has no reachable goal state from*
 489 *$abs(G, W_0)$, then no declared run of that pack configuration reaches φ_{goal} . Hence an impossi-*
 490 *ble verdict is never wrong relative to the pack, provided STEP-SIM and GOAL-SIM hold.*

491 *Proof.* We prove the contrapositive: if *some* declared run reaches the goal, then the abstract
 492 system reaches an abstract goal state. Suppose $C_0 \rightsquigarrow^* C$ with $goal_{\varphi_{goal}}(C)$. By Lemma 1,
 493 $abs(C_0) \rightsquigarrow^* abs(C)$, and by GOAL-SIM, $\hat{goal}(abs(C))$. Thus $abs(C)$ is an abstract goal state reach-
 494 able from $abs(C_0)$, contradicting the hypothesis. Hence no declared run reaches φ_{goal} : a verdict
 495 of *impossible* — read off exactly the premise, that the abstract search exhausts without hitting an
 496 abstract goal — is never wrong. Mechanized, axiom-free as a theorem schema parameterized by the
 497 two simulation hypotheses (the existential witness is $abs(C)$): $\square$

```

498   Theorem refutation_sound : forall s0,
499     (~ exists a, reach_astype (abs s0) a /\ agoal a) ->
500     (~ exists w, reach_cstep s0 w /\ cgoal w).
501     (* Print Assumptions refutation_sound. => Closed under the global
502       context *)

```

503 **Lemma 3** (The symbolic abstraction satisfies the hypotheses). *Let $abs(G, \langle B, N \rangle) = (G, \langle B, \psi \rangle)$ for*
 504 *any accumulated constraint $\psi \in \mathcal{L}$ with $N \models \psi$. Let an abstract step apply the same Add/Del update*
 505 *to B and replace ψ by the strongest postcondition of the numeric effect, and let $\hat{goal}(G, \langle B, \psi \rangle)$ hold*
 506 *when goal holds at control G and $\psi \wedge \varphi_{goal}$ is satisfiable under B . Then STEP-SIM and GOAL-SIM*
 507 *hold, and the widening $\psi \mapsto \top$ preserves both.*

508 *Proof.* GOAL-SIM: if $\langle B, N \rangle \models \varphi_{goal}$ and $N \models \psi$, then N itself witnesses satisfiability of $\psi \wedge \varphi_{goal}$
 509 under B . STEP-SIM: a concrete firing needs $\langle B, N \rangle \models pre_a$, so $\psi \wedge pre_a$ is satisfiable (N witnesses
 510 it) and the symbolic edge exists; the boolean update is identical on both sides, and the concrete
 511 successor satisfies the strongest postcondition by definition. Widening weakens ψ , which only adds
 512 abstract states and edges, so both properties are preserved (Theorem 2). Hence Theorem 1 applies to
 513 the search the implementation runs. This lemma is on paper; mechanizing the symbolic state is future
 514 work (Section 7). $\square$

515 **Theorem 2** (Tolerance soundness). *If a coarser over-approximation (more edges) cannot reach the*
 516 *goal, neither can any finer system. Hence coarsening α to tolerate more detail never produces a false*
 517 *refutation.*

518 *Proof.* Let $\rightsquigarrow_{fine}$ and $\rightsquigarrow_{coarse}$ be the two abstract step relations, with the coarsening an *over-*
 519 *approximation*: every fine step is also a coarse step, $x \rightsquigarrow_{fine} y \Rightarrow x \rightsquigarrow_{coarse} y$. Suppose, for
 520 contradiction, that the finer system reaches the goal: $s_0 \rightsquigarrow_{fine}^* a$ with $\hat{goal}(a)$. By Lemma 2 (with
 521 $\rightsquigarrow_1 = \rightsquigarrow_{fine}$, $\rightsquigarrow_2 = \rightsquigarrow_{coarse}$), $s_0 \rightsquigarrow_{coarse}^* a$; and $\hat{goal}(a)$ still holds. So the coarser system reaches
 522 a goal state, contradicting the hypothesis. Hence no coarser-refuted goal is reachable in any finer
 523 system: coarsening α only *adds* abstract edges, so it can never turn an unreachable goal into a
 524 reachable one — refutation is preserved. Mechanized as `tolerance_sound`, axiom-free. $\square$

525 **Theorem 3** (Capability monotonicity). *If $\Gamma \subseteq \Gamma'$ and Γ reaches φ_{goal} , then Γ' reaches φ_{goal} .*
 526 *Granting tools never makes an achievable goal impossible.*

527 *Proof.* Enlarging the capability context only *adds* guarded effects: since $\Gamma \subseteq \Gamma'$, every capability
 528 firing available under Γ is available under Γ' with the same guard and effect (WORLD-ACT in B.1
 529 quantifies over $a \in \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$), so every step of the Γ -transition relation is a step of

the Γ' -relation. Suppose Γ reaches the goal configuration C from C_0 . By Lemma 2, the same configuration is reachable under Γ' , and $goal_{\varphi_{goal}}(C)$ is unchanged because the goal predicate does not depend on Γ . Hence Γ' reaches φ_{goal} via the same witnessing run. The other premises of ACH are monotone in the same direction ($caps(G) \subseteq \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$ still holds, and realizability and conformance do not mention Γ beyond $a \in \text{dom}(\Gamma)$), so the whole judgment is preserved. The transport half is mechanized as `cap_monotone`, axiom-free, stated over an inclusion of step relations; the reduction of $\Gamma \subseteq \Gamma'$ to that inclusion is the WORLD-ACT argument above, on paper. $\square$

Concrete instance. The flight skill of Section 2, variant A, is mechanized as `FlightInstance`. We prove `flight_refuted` (no run reaches `booked` $\wedge$ `confirmation_sent`, since no step touches `conf`) and `booking_reachable` (a booking is still reachable). Both are axiom-free, witnessing non-vacuity of Theorem 1.

D.3 Operational correspondence: subject reduction and session fidelity

The direct judgment $G \vdash (\mathbb{M}; W)$ is not merely preserved by reduction: the session and its protocol step *in lockstep* over the labelled transition systems of Section B.2. This is the standard soundness backbone of a session calculus [13, 11], obtained here *directly*, with no projection and no merge. When actions by different participants change the world, we assume that their effects commute. Communication-only reductions satisfy this condition automatically. Goal markers require no additional condition because G-GOAL-F checks the goal in the world state where the step occurs.

Lemma 4 (Residual capability). *If $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, then $\Lambda \in \text{cap}(G)$.*

Lemma 5 (Participant agreement). *If $G \vdash (\mathbb{M}; W)$, then $\text{prt}(G) = \text{prt}(\mathbb{M})$.*

Both follow by induction on the transition or typing derivation, respectively; they are the capability-membership and participant invariants used in the interleaving cases below.

Theorem 4 (Subject reduction). *If $G \vdash (\mathbb{M}; W)$ and $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$, then $(G, W) \rightsquigarrow_{\Lambda} (G', W')$ for some G' with $G' \vdash (\mathbb{M}'; W')$.*

Theorem 5 (Session fidelity). *If $G \vdash (\mathbb{M}; W)$ and $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, then $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$ for some $\mathbb{M}'$ and W' with $G' \vdash (\mathbb{M}'; W')$.*

We prove both by induction on the typing derivation $G \vdash (\mathbb{M}; W)$, with a case analysis on the transition; the one auxiliary fact used throughout is that typing respects pointwise-equal sessions (`ctypes_ext`), which lets the interleaving cases reassociate independent updates.

Proof of Theorem 4 (subject reduction). By induction on $G \vdash (\mathbb{M}; W)$.

Case T-END ($G = \text{End}$, every role idle). Then $\mathbb{M}$ has no output or pending action, so no session transition exists and the claim holds vacuously.

Case T-GOAL ($G = \check{\varphi}.G_0$, with $W \models \varphi$ and $G_0 \vdash (\mathbb{M}; W)$, $\text{prt}(G_0) = \text{prt}(\mathbb{M})$). A marker is not a session construct, so the step $(\mathbb{M}, W) \rightarrow_{\Lambda} (\mathbb{M}', W')$ is a step of G_0 's session; the induction hypothesis gives $(G_0, W) \rightsquigarrow_{\Lambda} (G', W')$ with $G' \vdash (\mathbb{M}'; W')$. Since $W \models \varphi$, rule G-GOAL-F discharges the marker and performs the same step, $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda} (G', W')$; the residual typing is the one the induction hypothesis provides. No stability hypothesis is needed: the marker is checked at W , where the step happens.

Case T-COMM, with head $p \rightarrow q$ and branches G_i . Here $\mathbb{M}$ is $p[q! \dots] \parallel q[p? \dots] \parallel \mathbb{M}_0$, and each branch satisfies $G_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0; W)$. The transition is a communication $\Lambda = r \ell t$. Because p offers only outputs to q and q only inputs from p , the sender/receiver shapes force a dichotomy.

- *Head move* ($r = p$). Then $t = q$ and $\ell = \ell_k \in I$, and $\mathbb{M}' = p[P_k] \parallel q[Q_k] \parallel \mathbb{M}_0$. Rule G-COMM-E gives $(G, W) \rightsquigarrow_{p\ell_k q} (G_k, W)$, and $G_k \vdash (\mathbb{M}'; W)$ is exactly the k -th premise.
- *Interleaving move* ($r \neq p$). The shapes then force $r, t \notin \{p, q\}$: r is an output so $r \neq q$, and t is an input so $t \neq p$, while $t = q$ would force $r = p$. Since r, t lie in the bystanders $\mathbb{M}_0$, the *same* communication is enabled in each branch's configuration $p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0$ (the head updates do not touch r, t). The induction hypothesis at each i yields G'_i with

578 $(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W)$ and $G'_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}'_0; W)$, where $\mathbb{M}'_0$ is $\mathbb{M}_0$ after the
 579 bystander step. As $\text{prt}(\Lambda) = \{r, t\}$ is disjoint from $\{p, q\}$, rule G-COMM-I assembles
 580 $(G, W) \rightsquigarrow_{\Lambda} (p \rightarrow q: \{\ell_i, G'_i\}, W)$, and re-applying T-COMM (the sender/receiver at p, q
 581 are untouched, and each branch is retyped by the induction hypothesis, reassociating the
 582 independent updates via `ctypes_ext`) gives the residual typing.

583 Communications leave the world fixed, so no independence hypothesis is needed. For a world-
 584 changing action at the head (T-ACT), G-ACT-E matches and the residual G is typed at the *same*
 585 post-effect world the rule reaches — whichever successor world the session’s firing picks, the
 586 universal premise of T-ACT types the residual at it. Interleaving *two* actions requires their effects
 587 to commute ($\text{eff}_a \circ \text{eff}_b = \text{eff}_b \circ \text{eff}_a$ for disjoint a, b), a frame-independence side condition that
 588 participant-disjointness alone does not supply; under it the T-ACT interleaving case goes through
 589 identically. $\square$

590 *Proof of Theorem 5 (session fidelity).* By induction on $G \vdash (\mathbb{M}; W)$, inverting the global step
 591 $(G, W) \rightsquigarrow_{\Lambda} (G', W')$. T-END admits no global step in the correspondence, so the claim holds
 592 vacuously. For T-GOAL, the step is G-GOAL-F; its premise $(G_0, W) \rightsquigarrow_{\Lambda} (G', W')$ is realised
 593 by the induction hypothesis as a session step, which is the required one — the marker exists only
 594 on the type side and is discharged by the same transition. For T-COMM, the global step is either
 595 G-COMM-E — and the prescribed head communication $p \ell_k q$ is enabled by S-COMM, landing in
 596 the k -th premise — or G-COMM-I with $\text{prt}(\Lambda) \cap \{p, q\} = \emptyset$; then the induction hypothesis on any
 597 branch produces a bystander communication, whose endpoints lie outside $\{p, q\}$, so S-COMM fires it
 598 on $\mathbb{M}$ and T-COMM retypes the result. The action cases mirror those of Theorem 4. $\square$

599 **Mechanization.** Both theorems are mechanized axiom-free for the single-label communication
 600 fragment — the fragment in which bystander interleaving is unconditional — in `DirectTypingSR.v`,
 601 over labelled session and global transition systems, with no projection, merge, or local type:

```
602 Theorem subject_reduction : forall W G s s' L,
603   ctypes W G s -> lstep L s s' ->
604   exists G', gstep W L G G' /\ ctypes W G' s'.
605 Theorem session_fidelity : forall W G G' s L,
606   ctypes W G s -> gstep W L G G' ->
607   exists s', lstep L s s' /\ ctypes W G' s'.
608 (* Print Assumptions => Closed under the global context (both). *)
```

609 The head-move case for world-changing *actions* is additionally mechanized in `DirectTyping.v`
 610 (`type_directed_safety`, also `progress`). Goal labels appear in neither label alphabet: the
 611 paper’s correspondence uses the same fused goal rule (G-GOAL-F) as the Coq `GS_Goal`. Participant-
 612 matching side conditions, labelled action interleaving, and multi-branch choice are still absent from
 613 the mechanization. Completing a faithful mechanization of the paper rules remains work in Section 7.

614 **Concrete instance.** `HandoffInstance` mechanizes the planner/worker handoff of Section 1 on both
 615 sides. The *good* pairing — planner sends, worker delivers, goal checked — is proved `ctypes`-typed
 616 (`good_typed`) and its run is proved to reach a world satisfying the goal (`good_runs_to_goal`). The
 617 *bad* pairing — both roles start with an input, exactly the deadlock of Section 1 — is proved `Stuck`
 618 (`bad_stuck`), and the contrapositive of `progress` then gives, for free, that *no non-trivial global type*
 619 *can ever type it* (`bad_untypeable`): the rule rejects the classical deadlocking handoff without any
 620 projection ever being computed.

621 D.4 Incompleteness, by design

622 **Proposition 1** (Incompleteness). $\Gamma; G \models \Diamond_{\text{init}} \varphi_{\text{goal}}$ may hold while no concrete runtime execution
 623 reaches φ_{goal} : the witnessing abstract path can be spurious when α collapses a distinction that
 624 mattered. The system is sound for $\times$ and incomplete for $\checkmark$.

625 This is the price of decidable, tolerant over-approximation; tightening α trades tolerance for precision.
 626 The residue is confined to two edges, motivating the layered architecture:

- **Payload faithfulness** (bottom edge): a tool’s declared post-condition (“filters to fares < 500”) may be false — a runtime obligation for a deterministic monitor (Layer C), not a static one.
- **Intent fidelity** (top edge): the formalized goal may not capture what the user meant — irreducibly semantic, surfaced at the single compaction checkpoint for human review.

D.5 Decidability and the autonomy boundary

Theorem 6 (Termination and soundness of the widened search). *In the fragment with finitely many roles (static topology), regular global types represented by finite tail-recursive control graphs, decidable state logic $\mathcal{L}$, and $\mathcal{L}$ -definable effects, the widened symbolic search terminates on every pack. It decides the widened abstract judgment; for $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ its IMPOSSIBLE answer is sound (Theorem 1 with Lemma 3) and its ACHIEVABLE answer may be spurious (Proposition 1).*

Proof. The pack-level achievability judgment $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ (Section B.2) is an existential reachability query over configurations (G', W) ; we exhibit the terminating procedure and place its two answers.

The search space is finite. A configuration has two components. (i) *Control*, the residual global type G' reachable from G by $\rightsquigarrow$. Since G is a regular tree (tail-recursion: finitely many distinct subtrees, Section A.3) and each $\rightsquigarrow$ step either descends into a subtree (G-COMM-E, G-ACT-E) or erases a marker (G-GOAL-E), the set $\{G' : (G, _) \rightsquigarrow^* (G', _)\}$ is a subset of the finitely many subtrees of G , hence finite. (ii) *Symbolic world*. The boolean part is a valuation $B : \text{Pred} \rightarrow \mathbb{B}$ over the finitely many predicates named in the pack, so there are $2^{|\text{Pred}|}$ boolean states. The numeric part is summarized by an accumulated constraint $\psi \in \mathcal{L}$ (a conjunction of the guards taken and the Nd -postconditions assumed); on a control-graph back edge the reference procedure applies the widening $\psi \mapsto \top$, forgetting accumulated numeric constraints. This one-element numeric summary is an over-approximation and, together with the finite boolean valuations, leaves only finitely many symbolic worlds. The product with finite control is therefore finite.

Each edge is computable. Firing a capability requires checking $W \models pre_a$ and forming sp ; testing goal-satisfaction requires $W \models \varphi$. Both are entailment/satisfiability queries in quantifier-free linear integer arithmetic with uninterpreted predicates — a decidable theory, discharged by an SMT solver. Branch selection (REACH-OR) is a finite disjunction. Existence of an initial world is decided by the same SMT query over $init$; each satisfying symbolic initial valuation seeds the finite search.

Termination and correctness. Forward search (breadth-first over $\rightsquigarrow$) maintains a visited set of (control, symbolic-world) pairs; because that set is finite it adds each pair at most once, so the search halts. It answers ACHIEVABLE iff it enumerates a goal-satisfying configuration, and IMPOSSIBLE otherwise. Soundness of the IMPOSSIBLE verdict is Theorem 1 with Lemma 3; and widening only enlarges the reachable set (it weakens constraints, adding edges), so by Theorem 2 it never introduces a false refutation. Hence the search always terminates and decides the widened abstract judgment; IMPOSSIBLE transfers to the exact judgment, ACHIEVABLE does not (Proposition 1). $\square$

Why not exact decidability. With unbounded integers, updates $x := x \pm 1$, and guards $x = 0, x \geq 1$ under recursive control, packs encode two-counter machines, so the exact judgment $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ is undecidable even with a fixed set of participants. The widening is what buys termination; the price is the spurious-ACHIEVABLE residue of Proposition 1.

Proposition 2 (Undecidability in a dynamic-topology extension). *Consider the extension G^+ of the grammar with add r. G, which creates (adds) a new participant during execution: the new participant runs a declared finite behaviour template and brings one fresh numeric variable, initialized by the step that adds it. With an unbounded role population and dynamic channels, achievability in G^+ is undecidable — and no widening restores termination, since the reachable control space itself becomes infinite.*

Proof sketch. By reduction from the control-state reachability problem for communicating finite-state machines (CFSMs), which is undecidable [15]. Fix a CFSM instance: finite-control machines M_1, M_2 exchanging messages over unbounded FIFO channels, and a target control state q_f of M_1 ; deciding whether a run reaches q_f is undecidable.

We construct a skill whose goal is achievable iff the CFSM reaches q_f . Each machine M_i becomes a role whose process follows its control graph. When a machine sends message m , the process adds a new participant carrying m . When a machine receives m , it synchronizes with the oldest outstanding participant for that channel. This unbounded set of participants represents the unbounded FIFO queue. The message order is enforced using sequence numbers: each new participant stores the current send counter, and a receive is allowed only when that number matches the receive counter. This requires a fresh variable for each participant, which is available in G^+ but not in the base fragment with a fixed Var.

The *Asg/Nd* machinery of Section A.2 supplies the counter updates. A boolean predicate $\text{at_}q_f$ is established by the single capability guarding M_1 's entry into q_f , and we set $\varphi_{goal} = \text{at_}q_f$. By construction a run of the skill reaches a state satisfying φ_{goal} iff the CFSM has a run reaching q_f : the courier discipline then puts the skill's reachable configurations in correspondence with the CFSM's channel contents and control states. The construction is a computable map from CFSM instances to packs, so a decision procedure for achievability would decide CFSM control-state reachability — impossible. Hence achievability is undecidable once new participants may be added during execution. (The finiteness argument of Theorem 6 breaks at exactly the point this reduction exploits: an unbounded number of couriers makes the set of reachable controls infinite.) $\square$

The two results identify a concrete boundary: when the set of participants is finite and fixed, the widened search terminates and its *impossible* verdict is sound. If an unbounded number of new participants can be added during execution, no such finite search can decide the problem in general. This result concerns only the number of participants; it does not mean that all forms of agent autonomy are undecidable. The implementation returns UNKNOWN when a pack violates the static-topology side condition of Theorem 6, unless an independent capability or establisher refutation has already proved it impossible.

D.6 The partition of obligations

Corollary 1. *The architecture separates “will the goal be achieved” into four obligations on disjoint substrates:*

Concern	Where	Mechanism
goal achievability	static, terminating	Coq specification; deterministic search; no LLM
per-step safety / least privilege	Layer B (future)	same engine, $\square$ + permission tiers
payload faithfulness	Layer C (future), run-time	deterministic monitors with blame
intent fidelity	top edge, synthesis	one human checkpoint

D.7 Verifiable feedback beyond a scalar reward

The verdict is a heterogeneous verification signal rather than a scalar score. An IMPOSSIBLE result identifies which premise failed and carries a blocking frontier; ACHIEVABLE carries a witness path and explicitly retains the two deferred obligations; UNKNOWN is an abstention outside the decidable fragment, not a refutation. These structured outcomes can serve as cheap negative labels for automated skill or prompt search without an LLM judge, while the intent-fidelity checkpoint keeps the irreducible semantic decision with a human. We treat such search as an application of the verifier, not as a contribution evaluated here.

E Token economics of pre-execution refutation

The practical case for deciding achievability *before* execution is an economic one, so we state it in tokens rather than adjectives. Two asymmetries carry it. First, no model sits in the trusted path: the schema gate, direct conformance and Z3 reachability spend **zero tokens**, and the deterministic front-end spends zero as well. Only the optional LLM compaction spends tokens, and it is paid *once per skill version*. Second, an agent turn re-sends its whole context: writing S for the harness prompt, K for the loaded skill, g for the mean tokens appended per turn and o for output per turn, a run of T turns reads $T(S+K) + gT(T-1)/2$ input and To output tokens. Compaction is linear in the skill and paid once; a doomed run is *quadratic* in its turns and recurs on every invocation.

Runtime waste is necessarily a model — it prices a run that, if the refutation is correct, never happens — so we publish the model rather than a single number. Each refutation reason carries a (low, typical, high) band for how long an agent flails before that failure stops it, and how many agents are billed. With conservative defaults ($S=2500$, $g=700$, $o=250$) and a median real consumer skill ($K \approx 1.7K$ tokens), compaction costs $\sim 2.8K$ tokens against the following avoided waste per prevented invocation:

	Verdict reason	turns (lo–typ–hi)	waste/run	leverage
	MISSING_CAPABILITY	3–8–14, 1 agent	50 240	18×
	GOAL_UNSAT	8–18–30, 1 agent	176 040	63×
	NON_CONFORMANT	6–15–30, 2 agents	261 900	94×
	NON_PROJECTABLE	6–15–30, 2 agents	261 900	94×
	BLOCKED_GUARD	12–25–50, 1 agent	305 750	110×

The ordering is the robust part, and it is the expected one. `MISSING_CAPABILITY` is cheapest at run time: the agent calls a tool that is not there and learns immediately, and most of the cost is the improvisation that follows. `BLOCKED_GUARD` is dearest, because a precondition no run can satisfy is precisely the case an agent cannot learn from — it retries to the harness turn cap. The two multi-party reasons bill two contexts. `GOAL_UNSAT` additionally carries a cost the token bill does not show: a run whose goal conjunct has no establisher can terminate *believing* it succeeded. `DYNAMIC_TOPOLOGY` claims no savings at all, because an abstention prevents nothing.

Over the 15-spec corpus the seven structural refutations cost 12.2K tokens of compaction and avoid $\sim 1.07M$ tokens per round of invocations (band 0.28M–3.15M): $\sim 87\times$ leverage, or nothing at all on the deterministic path. Break-even therefore falls *inside* the first prevented run in every mode — between 0.9% and 5.5% of it. The honest denominator for a healthy skill, where the check buys nothing and still costs something, is 2.9% of one successful run over the corpus and 4.0% for a median real skill; with the deterministic front-end, zero. Prompt caching changes the price of the wasted tokens, not their number. Halving every waste default leaves leverage at $9\times$ – $55\times$ and the ordering unchanged. The model, its defaults and their justification are in the artifact (`skillc.tokens`, `skillc.cost`); the Coq development costs no tokens and is amortized over every skill ever checked.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Question: Does the paper discuss the limitations of the work performed by the authors?

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 768 perimental results of the paper to the extent that it affects the main claims and/or conclusions
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